UDCT v7.14: Exploratory QUMOND Hessian and Phase-Space Consistency Tests (Steps 01–50)

Comprehensive research paper and source-by-source technical commentary | First public version | 26 October 2026

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Status: Pre-preregistration exploratory record. Neither peer reviewed nor a direct observational validation. The DOI remains reserved until Zenodo publication.

Abstract

This extended manuscript documents a 50-stage computational program progressing from Hydrus I-motivated observational sensitivity checks to three-dimensional QUMOND field/Hessian audits, Jeans-equation consistency tests, and isolated spherical distribution-function (DF) inversions. Its purpose is to preserve assumptions, numerical artifacts, mathematical arguments, test boundaries, and failure modes rather than announce a new gravitational law. The later branch employs a Plummer gravitating source and the tracer family rho*(r)=rho0[1+(r/a)^p]^(-5/p). Under specific central scaling and Abel-inversion regularity conditions, the leading asymptotic DF coefficient changes sign across p=3/2; at exact equality it cancels and cannot establish the full DF sign. Archived Osipkov–Merritt (OM) calculations report a finite-window local tangency r_a/a=0.46646678739601644 for p=1.5, with energy-label r/a=0.47409755283222904. Neither a finite-window root nor a negative DF in this ansatz is a global MOND falsification. The early 3D Hessian computations are distinct from the spherical DF branch and do not by themselves implement the generalized GQUMOND field equations. The companion archive preserves original source code, CSV/JSON records and per-step instructions. No full 50-stage independent replication or individual-star Hydrus I likelihood fit is claimed.

Reader guide and central conclusions

• The first part presents physical motivation, equations, inversion assumptions, interpretation and limits; the second part provides the 50-step provenance narrative and detailed file-level audit.

• The Hessian, Jeans second moment, phase-space DF, and dynamical stability are separate mathematical tests; no inference between them is automatic.

• The local p=1.5 coefficient cancellation is a particularly important unresolved equality case, not a proven global threshold.

• The Step 50 anisotropy value is local to the declared scan window; all-energy admissibility requires further tests.

• Figures are redrawn from archived CSV outputs. This document does not represent a new independent computational campaign.

1. Scientific lineage and research question

Bekenstein and Milgrom (1984) introduced AQUAL as a nonlinear modified-gravity theory. Milgrom (2010) formulated QUMOND as a quasilinear construction. Milgrom (2023) developed GQUMOND actions depending on higher spatial derivatives of the Newtonian potential. This 2023 article is crucial prior art for Hessian-sensitive modifications; a diagnostic evaluation of a Hessian tensor is not equivalent to deriving or solving the full action-dependent GQUMOND field equations. The current 3D Hessian branch is described as a diagnostic and numerical validation exercise, while the later spherical DF branch is tied to a restricted QUMOND potential.

The research question is not which gravity camp wins. It is whether specified combinations of gravitational potential, stellar tracer density and orbital anisotropy can be represented by a physically admissible, nonnegative equilibrium distribution function. A failure excludes that particular model/ansatz under the stated assumptions, not every model in its parent theory.

2. Data provenance and the two research branches

The supplied primary sources are 7.14 Hessian 1~25.zip (80 top-level entries) and 7.14 Hessian 26~50.zip (25 nested step packages). The first archive contains a mixture of early experimental scripts, data tables and numbered later packages; some early files cannot be uniquely assigned to Steps 01–15. The second archive has explicit Steps 26–50. Source-file names are kept rather than retrospectively inventing a complete chronology. ZIP integrity, ability to execute Python, numerical reproducibility, correct equations and observational agreement are different levels of evidence.

Early work explored membership, foreground, metallicity, velocity gradients, uncertainty, baseline Newtonian comparisons and other Hydrus I-motivated issues. Steps 16–25 focused on 3D fields and Hessian convergence, boundaries and rotations. Steps 26–38 transitioned through Jeans and DF inversion. Steps 39–44 examined critical exponents, independent quadrature, forward reconstruction and tail sensitivity. Steps 45–50 explored OM anisotropy, sign maps, zero contours, lobe collapse and local tangency.

3. Field models and the 3D Hessian

For a Newtonian potential Phi_N, the Hessian is H_ij = partial_i partial_j Phi_N. Under an orthogonal rotation R, the exact tensor transforms H -> R H R^T, preserving trace and eigenvalues. Discrete Cartesian finite differences need not preserve this identity exactly; convergence and rotation audits quantify numerical artifacts.
Boundary-condition effects, finite-box size and external-field surrogates must be distinguished from genuinely covariant field equations. Eigenvalue signs can indicate local tidal geometry, but do not establish the sign of a stellar distribution function.

In conventional QUMOND, a Newtonian Poisson solve supplies the Newtonian field, and a second Poisson solve obtains the modified potential using a nonlinear source built from the Newtonian acceleration. GQUMOND permits additional higher-derivative dependence in its action, changing the effective source. One cannot infer implementation of a GQUMOND Lagrangian merely from Hessian arrays being computed. The current release retains the distinction as a reproducibility and prior-art safeguard.

4. Spherical source, tracer and energy variables

The late DF branch assumes an isolated spherical Plummer-type gravitating source and a separately specified tracer family rho_star(r) = rho0 [1+(r/a)^p]^(-5/p). The source mass density and tracer density need not coincide. The source length scale quoted in the earlier manuscript is a=53.3 pc, which is an archived modeling choice rather than a universal galaxy constant. The potential convention matters: DeltaPhi=Phi(r)-Phi(0) increases outward, whereas some standard Eddington presentations use the relative potential Psi=-Phi+Phi(infinity). Equations must not mix these conventions without changing integration limits and boundary terms.

For a cored spherical Newtonian source, the inner Newtonian acceleration scales as g_N~r. In the assumed deep-MOND central regime, g~sqrt(a0*g_N)~r^(1/2). Therefore DeltaPhi~r^(3/2). The tracer expansion at small r is rho_star(r)=rho0[1-(5/p)(r/a)^p+O(r^(2p))]. Eliminating r gives rho(DeltaPhi)=rho0-D(DeltaPhi)^s+..., with D>0 and s=2p/3. Each proportionality assumes the stated central regime and smoothness; an external field or another central mass law can change the exponent.

5. Eddington/Abel inversion and positivity

An isotropic equilibrium DF f(E) produces a density by integration over allowed velocities. With absolute potential and E=v^2/2+Phi(r), rho(Phi)=4*pi*sqrt(2)*Integral_{Phi}^{infinity} f(E)*sqrt(E-Phi) dE for the appropriate convention. Inverting this Abel-type transform involves differentiating the density as a function of potential and accounting for endpoints. For the conventional relative-potential convention, f(Eps)=(1/(sqrt(8)*pi^2))[Integral_0^Eps dPsi (d^2rho/$dPsi^2$)/sqrt(Eps-Psi)+(1/sqrt(Eps))(drho/dPsi)|Psi=0]. The exact signs and boundaries depend on which energy convention is chosen.

A negative DF at an accessible phase-space point contradicts a nonnegative distribution for the specified equilibrium density/potential pair and chosen inversion. It is not necessarily a negative local velocity dispersion, which is a second moment of a distribution, nor proof of linear instability. Numerical inversion can also produce false negative lobes if energy sampling, truncation, interpolation, endpoint subtraction or derivative regularization is inaccurate; independent quadrature and forward reconstruction are therefore essential controls.

6. Central asymptotics and the critical exponent

Let s=2p/3 and rho(DeltaPhi)=rho0-D(DeltaPhi)^s+..., D>0. The archived derivation records a leading singular contribution proportional to -D*s*(s-1)*Beta(1/2,3/2-s)*E^(s-3/2), subject to convergence and dominance assumptions. For 0<s<3/2 the beta factor is positive. Accordingly the displayed coefficient is positive when 0<p<1.5 and negative when 1.5<p<2.25. These statements concern the leading singular term in the stated range, not every energy of the complete inverted DF.

At p=1.5, s=1 and the factor (s-1) vanishes. This does NOT imply f=0, positivity, or negativity: the next nonzero term, regular contributions, and any boundary term can dominate. Near p=2.25 the beta-function expression reaches a convergence boundary; extension outside the stated interval requires separate asymptotics. A rigorous equality-case conclusion requires an independent derivation of the next-order potential and tracer expansions, inversion of those terms, and a numerical check at decreasing energy scales.

7. Osipkov–Merritt orbital anisotropy

For the standard radially anisotropic OM family, beta(r)=r^2/(r^2+r_a^2), and the augmented tracer density is rho_aug(r)=(1+r^2/r_a^2)rho_star(r). The DF depends on Q=Eps-L^2/(2r_a^2), subject to the selected potential convention. Because the Abel inversion is linear in augmented density, f(Q;r_a)=f_iso(Q)+B(Q)/(r_a/a)^2 for a fixed potential and tracer. This decomposition is useful for verifying contour roots and numerical monotonicity, but the positivity domain remains a function of all accessible Q. A tangency of a sampled negative lobe is not a proof of a global critical anisotropy radius.

8. Archived finite-window sign boundaries

The Step 48 example p=1.5, r_a/a=0.4 reports two zero labels r/a≈0.22370244 and 0.76696971 and a normalized minimum DF ≈-0.0008538448. These r/a values are energy labels induced by the monotonic spherical potential, not literal boundaries of a spatial shell of unphysical stars. Step 50 scans the specified window 0.25≤r/a≤0.75 and reports local tangency r_a/a=0.46646678739601644 at label r/a=0.47409755283222904. Nearby archived results: p=1.49 -> r_a/a=0.462739644607, label r/a=0.476369715642; p=1.501 -> r_a/a=0.466841165090, label r/a=0.473866183835.

The finite-difference derivative near the reported tangency is close to zero, but a small residual is a local numerical diagnostic. Comparisons of quadrature orders test internal numerical stability; they do not independently establish that the potential, tracer, DF kernel or model assumptions are correct.

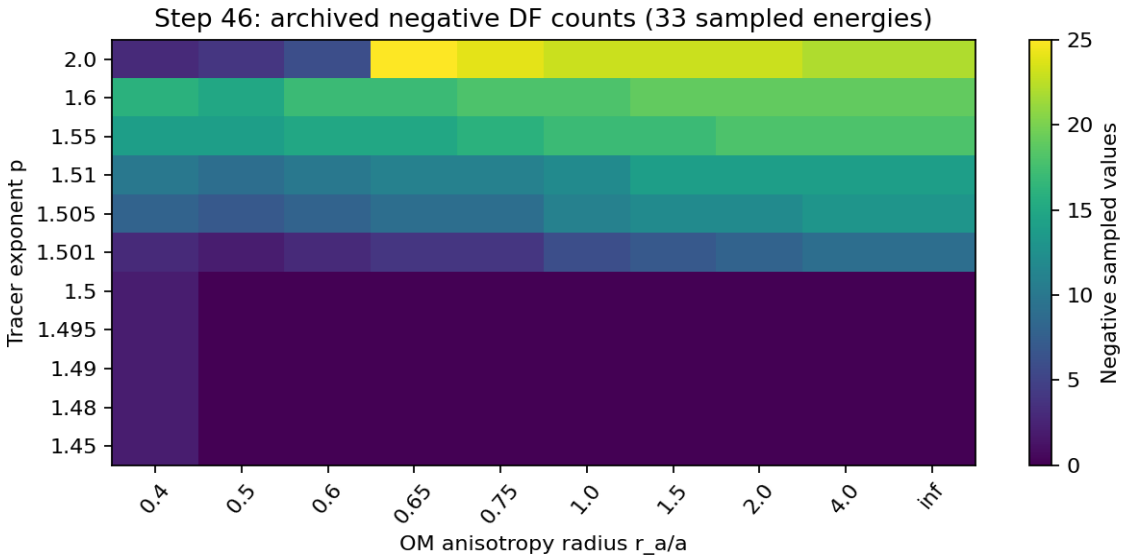


Figure 1. Step 46 archived discrete negative-DF count map; not an all-energy positivity certificate.

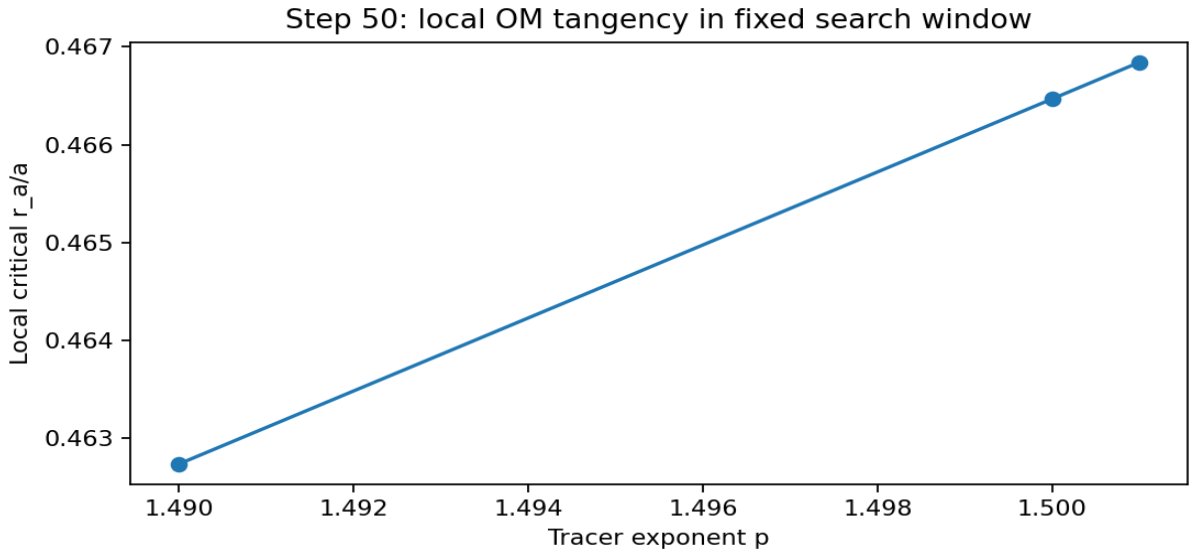


Figure 2. Step 50 local tangency values for nearby tracer exponents.

9. Jeans equations, Hessian, DF and stability: non-equivalent tests

The spherical Jeans equation constrains the tracer-weighted radial second moment: d(rho*sigma_r²)/dr + 2*beta*rho*sigma_r²/r = -rho*dPhi/dr. Solutions depend on boundary conditions, density and anisotropy; a negative formal second moment is a different inconsistency from a negative inverted DF. A Hessian eigenvalue describes curvature of the gravitational potential. A nonnegative DF is a stronger equilibrium phase-space requirement under a chosen ansatz. Dynamical stability requires additional perturbative or nonlinear time-evolution analysis. No one of these four signs can be substituted for another.

10. Observational motivation versus direct Hydrus I inference

Hydrus I observational work motivates scales and dwarf-galaxy stress tests, but using observation-informed model inputs is not equivalent to fitting the full individual-star line-of-sight velocity catalogue. A direct analysis needs per-star velocities, measurement errors, membership probabilities, foreground distributions, binary-star contamination, selection functions, distance and systemic velocity nuisance parameters, and likelihood comparisons across gravitational models. The present late spherical DF calculations are not a star-by-star fit. They are not a full 3D external-field DF inversion either.

11. Numerical credibility and audit tiers

Tier 1 — provenance: original ZIP, code and output files retained byte-for-byte, with checksums.

Tier 2 — execution: source script runs in a documented environment without failure.

Tier 3 — repeatability: regenerated CSV/JSON agree within stated tolerances or byte-for-byte.

Tier 4 — independent numerical method: distinct integration, interpolation and endpoint treatment agree.

Tier 5 — mathematical verification: action, field equation, potential and inversion are derived consistently.

Tier 6 — empirical validation: held-out stellar data are fitted under preregistered criteria and compared with alternatives.

Earlier internal records reported selected reexecution of five scripts (Steps 39, 42, 46, 47 and 50) and matching 13 CSV/JSON outputs. The current expanded manuscript audits archived source content and includes original files; it does not newly establish independent execution of all 50 steps. Reproducing a result with the same implementation does not guarantee independent mathematical correctness.

12. Detailed archived source inventory: early Steps 01–25

The following list preserves filenames exactly as supplied. It is not a verified one-to-one Step 01–25 chronology. A few entries may be numbered subpackages; early scripts are otherwise grouped by topic. Original files remain in the bundled ZIP.

audit_hydrus_20261026_021810_002.py | audit_hydrus_20261026_021854_287.py | bootstrap_replicates_20261026_021942_229.csv | bootstrap_stability_20261026_021944_811.py | bootstrap_summary_20261026_021947_149.json | center_anisotropy_20261026_022038_265.py | comparison_20261026_021957_126.csv | comparison_20261026_022211_162.csv

data_audit_20261026_021847_544.json | data_audit_20261026_021849_724.csv | efe_sensitivity_20261026_022224_937.csv | efe_surrogate_20261026_022226_922.py | exploratory_fit_20261026_021904_415.json | exploratory_fit_20261026_021906_264.py | fixed_mass_profile_20261026_022053_021.csv | gradient_sensitivity_20261026_021911_953.json

gradient_sensitivity_20261026_021913_880.py | J_MNRAS_479_5343_20261026_021817_471.mht | J_MNRAS_479_5343_20261026_021825_570.mht | J_MNRAS_479_5343_20261026_021900_639.mht | joint_membership_20261026_022123_743.csv | joint_velocity_metallicity_20261026_022126_149.py | membership_sensitivity_20261026_022137_636.csv | mixture_fit_20261026_021928_530.py

mixture_membership_diagnostics_20261026_021930_750.csv | mixture_results_20261026_021933_288.json | newtonian_jeans_baseline_20261026_022012_283.py | newtonian_jeans_results_20261026_022014_583.json | om_comparison_20261026_022151_944.csv | om_df_test_20261026_022155_524.py | README_20261026_021814_560.md | README_20261026_021851_500.md

README_20261026_022016_760.md | README_20261026_022024_698.md | README_20261026_022040_411.md | README_20261026_022140_172.md | results_20261026_022042_528.csv | results_20261026_022128_195.json | scale_distance_results_20261026_022026_708.csv | scale_distance_sensitivity_20261026_022028_496.py

sensitivity_20261026_022001_190.py | spatial_sensitivity_20261026_022142_414.csv | spherical_mond_jeans_20261026_022213_884.py | stellar_mass_comparison_20261026_022107_024.py | stellar_mass_diagnostic_20261026_022055_176.py | stellar_only_grid_20261026_022110_321.csv | summary_20261026_022003_389.json | summary_20261026_022044_255.json

summary_20261026_022058_322.json | summary_20261026_022112_149.json | summary_20261026_022159_098.json | summary_20261026_022216_139.json | summary_20261026_022230_319.json | table2_20261026_021828_693.txt |

table2_20261026_021903_065.txt | UDCT_v7_14_Step16_REDOWNLOAD_20261024_20261026_022240_494.zip

UDCT_v7_14_Step17_Boundary_Hessian_Audit_20261026_022248_250.zip |
UDCT_v7_14_Step18_Analytic_Hessian_Benchmark_20261026_022254_170.zip |
UDCT_v7_14_Step19_Convergence_Audit_20261026_022302_780.zip |
UDCT_v7_14_Step20_High_Resolution_Hessian_Audit_20261026_022308_779.zip |
UDCT_v7_14_Step21_Convergence_Reaudit_20261026_022315_272.zip |
UDCT_v7_14_Step22_External_Field_Boundary_20261024_20261026_022326_711.zip |
UDCT_v7_14_Step22_External_Field_Boundary_20261024_20261026_022507_212.zip |
UDCT_v7_14_Step23_3D_Rotation_Covariance_20261024_20261026_022524_581.zip

UDCT_v7_14_Step24_Oblique_Rotation_20261024_20261026_022530_355.zip |
UDCT_v7_14_Step25_3D_Oblique_Hessian_20261024_20261026_022537_848.zip |
UDCT_v7.14_Bootstrap_Stability_Exploratory_20261026_021949_596.zip |
UDCT_v7.14_Center_Anisotropy_Exploratory_20261026_022046_326.zip |
UDCT_v7.14_Exploratory_Mixture_Audit_20261026_021935_461.zip |
UDCT_v7.14_Fixed_Mass_Profile_Exploratory_20261026_022100_563.zip |
UDCT_v7.14_Foreground_Error_Sensitivity_Exploratory_20261026_022005_783.zip |
UDCT_v7.14_Gradient_Exploratory_20261026_021917_062.zip

UDCT_v7.14_Joint_Chemo_Kinematic_Exploratory_20261026_022130_572.zip |
UDCT_v7.14_Newtonian_Jeans_Exploratory_20261026_022019_017.zip |
UDCT_v7.14_Newtonian_Scale_Distance_Sensitivity_20261026_022030_550.zip |
UDCT_v7.14_Physical_Anisotropy_DF_Exploratory_20261026_022202_493.zip |
UDCT_v7.14_Scalar_EFE_Sensitivity_Exploratory_20261026_022233_061.zip |
UDCT_v7.14_Spatial_Membership_Sensitivity_20261026_022145_917.zip |
UDCT_v7.14_Spherical_MOND_Newtonian_Exploratory_20261026_022218_458.zip |
UDCT_v7.14_Stellar_Luminosity_Newtonian_Exploratory_20261026_022114_139.zip

13. Detailed stage-by-stage technical archive: Steps 26–50

For each numbered package below, the section records the source files and excerpts of the archived README or summary. Excerpts are evidence of the author-supplied research record, not independent certification. Original README, Python and machine-readable results are available in the accompanying ZIP. Long README files are excerpted rather than silently rewritten.

Step 26 — archived source and numerical evidence

Files (7): UDCT_v7_14_Step26_Jeans_3D_Link/3d_force_crosscheck.csv, UDCT_v7_14_Step26_Jeans_3D_Link/README.md, UDCT_v7_14_Step26_Jeans_3D_Link/jeans_tail_convergence.csv, UDCT_v7_14_Step26_Jeans_3D_Link/requirements.txt, UDCT_v7_14_Step26_Jeans_3D_Link/results.json, UDCT_v7_14_Step26_Jeans_3D_Link/step24_reference.py, UDCT_v7_14_Step26_Jeans_3D_Link/step26.py

Archived README synopsis: UDCT v7.14 Step 26 — 3D Force / Spherical Jeans Bridge PRE-PREREGISTRATION EXPLORATORY; no new theory claim. Run python step26.py (numpy, scipy). step24_reference.py is copied unchanged from Step 25's reference implementation. The code compares two new 3D finite-box QUMOND solutions (65^3, 97^3, 640 pc cube, zero external field, zero Dirichlet boundary) with the isolated spherical QUMOND analytic acceleration at Plummer scale a=53.3 pc. The finite-box boundary is not certified as isolated. It separately integrates the spherical isotropic Jeans equation for a Plummer tracer with Newtonian stellar-only and isolated spherical QUMOND acceleration. Newtonian analytic radial sigma_r^2(a)=GM/(6 sqrt(2) a) provides a check. Radial dispersion is not projected LOS dispersion. The spherical Jeans integration does not incorporate the 3D computed acceleration directly; the 3D force calculation is a local cross-check only. This is not a 3D Jeans solution and not an observed-star fit. jeans_tail_convergence.csv varies outer radius 1–30 kpc and integration spacing 1 to 0.5 pc. Values near the cutoff are artificial zero-boundary values. 3d_force_crosscheck.csv compares acceleration and reports Hessian eigenvalues, which must NOT be conflated with negative velocity dispersion. results.json preserves constants, output, limitations. Known limitations: spherical isotropy beta=0; no EFE; no selection-function model; no membership likelihood; no Hyd [excerpt]

Archived machine-readable result excerpt: {"status": "EXPLORATORY; no preregistration, no observed-star likelihood", "constants": {"G": 0.00430091, "a_pc": 53.3, "a0_in_(km/s)^2_per_pc": 0.0037028130977896406, "stellar_mass_Msun": 32731.808703363735}, "newtonian_radial_sigma2_analytic_at_a": 0.3112697391849166, "tail_convergence": [{"rmax_pc": 1000, "dr_pc": 1.0, "newton_sig2_at_a": 0.3113547337737417, "qmond_sig2_at_a": 0.38088414542196025, "qmond_sig2_center": 0.5405923271664871, "min_newton_sig2": 0.0863003830568652, "min_qmond_sig2": 0.1947822905281563}, {"rmax_pc": 1000,

"dr_pc": 0.5, "newton_sig2_at_a": 0.3112910466909195, "qmond_sig2_at_a": 0.3808103036488286, "qmond_sig2_center": 0.5408719752360824, "min_newton_sig2": 0.08629717570468, "min_qmond_sig2": 0.19477679272323145}, {"rmax_pc": 3000, "dr_pc": 1.0, "newton_sig2_at_a": 0.3113547903189382, "qmond_sig2_at_a": 0.38088452257866057, "qmond_sig2_center": 0.5405923938235 [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 27 — archived source and numerical evidence

Files (5): README.md, jeans_anisotropy_grid.csv, requirements.txt, results.json, step27.py

Archived README synopsis: UDCT v7.14 — Step 27: Spherical Jeans anisotropy audit Status: pre-preregistration exploratory. Not an observational validation or 3D Jeans solution. Run: python step27.py (requires numpy and scipy). Reproduces jeans_anisotropy_grid.csv and results.json . 48 numerical integrations: Newtonian vs isolated spherical QUMOND, constant beta -0.5, 0, 0.25, 0.5 and Osipkov–Merritt r_a/a 0.5, 1, 2, 5; outer integration limits 3000, 10000, 30000 pc. Grid spacing 0.5 pc. Radial sigma^2 is reported only at 0.5a, a, 2a, 5a (a=53.3 pc). Newtonian beta=0 analytic control at r=a gives GM/(6 sqrt(2) a). Important: Plummer has a core. Constant beta > 0 is not automatically a physically consistent distribution function; central cusp-anisotropy constraints must be tested. Osipkov–Merritt models also need distribution-function positivity checks. Positivity of Jeans sigma^2 alone is necessary but insufficient. No line-of-sight projection, actual individual Hydrus I star likelihood, external field, 3D force integration, or v7.13 B01–B24 replay is included. Previous Step26 is a distinct 3D force cross-check; this step is a spherical Jeans anisotropy extension.

Archived machine-readable result excerpt: {"scope": "Spherical Plummer tracer; Newtonian and isolated spherical QUMOND analytic radial force; exploratory anisotropic Jeans integration", "constants": {"G": 0.00430091, "a_pc": 53.3, "a0_km2_s2_pc": 0.0037028130977896406, "stellar_mass_Msun": 32731.808703363735}, "runs": 48, "newton_isotropic_at_a": 0.31129110330481574, "newton_isotropic_analytic_at_a": 0.3112697391849166, "newton_isotropic_relative_error": 6.863538985535378e-05, "all_sampled_sigma2_positive": true, "cautions": ["Not a direct 3D Jeans solution: spherical radial-force input only", "No external-field effect and no Hydrus I individual-star LOS fit", "Jeans sigma2 positivity does not prove a nonnegative phase-space distribution function", "Constant beta=0.5 can violate central cusp-anisotropy consistency for cored Plummer; do not call physically admissible", "Osipkov-Merritt choices also require DF consistency tests", [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 28 — archived source and numerical evidence

Files (5): README.md, om_distribution_positivity.csv, requirements.txt, results.json, step28.py

Archived README synopsis: UDCT v7.14 — Step 28: Phase-space DF positivity Status: Exploratory, before preregistration. This is a Newtonian Plummer Osipkov–Merritt benchmark, not a QUMOND distribution-function inversion, a 3D Jeans solution, or a fit to individual Hydrus I stars. Reproduce python step28.py with Python, numpy and scipy installed. This creates om_distribution_positivity.csv and results.json . Method For a self-consistent Newtonian Plummer sphere with relative potential Psi(r)=GM/sqrt(r^2+a^2) and OM augmented density rho_Q(r)=(1+r^2/ra^2)rho(r) , define u=Q/Psi0 , q=(a/ra)^2 , Psi0=GM/a . Then rho_Q/rho0=(1-q)u^5+q u^3 . Eddington inversion yields f(Q) = rho0/(sqrt(8) pi^2 Psi0^(3/2)) [(1-q) (128/7) u^(7/2) + 8 q u^(3/2)] . The nonnegative DF criterion for all 0<Q<=Psi0 is exactly q<=16/9 , i.e. ra/a>=3/4 . The script checks 2001 energies and independently reconstructs the augmented density by quadrature at five potential values for each model. Negative DF at ra/a=0.5 and positive DF at ra/a>=0.75 are not new physical discoveries; this is a code and physical-admissibility benchmark. For a cored Plummer tracer, the central cusp-slope anisotropy theorem excludes constant positive central beta (gamma0=0 implies beta0<=0) under its assumptions. Tangential constant beta values are not certified by this test. Interpretation and limitations - The Step27 isolated spherical QUMOND Jeans calculations are not validated as phase-space consistent [excerpt]

Archived machine-readable result excerpt: {"title": "UDCT v7.14 Step28 Newtonian Plummer OM DF admissibility audit", "scope": "Self-consistent isotropic/Osipkov-Merritt Newtonian Plummer only; analytic Eddington inversion plus numerical density reconstruction", "formula": "f(Q)=[rho0/(sqrt(8)*pi^2*Psi0^(3/2))]*[(1-q)*(128/7)*u^(7/2)+8*q*u^(3/2)], u=Q/Psi0, q=(a/ra)^2", "analytic_threshold": "ra/a >= 3/4 for all nonnegative f(Q), 0<Q<=Psi0", "samples": [{"ra_over_a": 0.5, "q": 4.0, "min_fscaled": -22.857142857142854, "fscaled_at_Qmax": -22.857142857142854, "negative_samples": 473, "positive_all_energies": false, "zero_crossing_Q_over_Psi0": 0.7637626158259734, "max_density_reconstruction_abs_error": 3.97326616052851e-12}, {"ra_over_a": 0.7, "q": 2.0408163265306127, "min_fscaled": -2.7055393586005856, "fscaled_at_Qmax": -2.7055393586005856, "negative_samples": 148, "positive_all_energies": false, "zero_crossing_Q_over_Psi0": 0.926 [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 29 — archived source and numerical evidence

Files (7): README.md, necessary_condition.csv, outer_reference_sensitivity.csv, potential_tail.csv, requirements.txt, results.json, step29.py

Archived README synopsis: UDCT v7.14 — Step 29: QUMOND OM DF admissibility preflight Status: exploratory, before preregistration. This package DOES NOT establish QUMOND distribution-function positivity. Reproduce pip install -r requirements.txt then python step29.py . Outputs: necessary_condition.csv , potential_tail.csv , outer_reference_sensitivity.csv , and results.json . Model and equations Spherical Plummer tracer and source, scale a=53.3 pc, luminosity 6546 Lsun, M/L=5 -> stellar mass M=32730 Msun. Isolated QUMOND with simple nu function nu(y)=0.5+sqrt(0.25+1/y) and a0=1.2e-10 m/s^2. The acceleration is g_Q(r)=nu(g_N/a0) g_N where g_N=GM r/(r²+a²)^(3/2) . This spherical algebraic expression is valid in the isolated spherical limit; it is NOT the nonspherical 3D external-field solution. OM augmented tracer density rho_aug(r)=[1+(r/ra)^2]rho(r) . A nonnegative OM DF implies d rho_aug/d Psi >= 0 for the usual finite-escape inversion (and, more generally, monotonicity of the corresponding velocity integral). Since dPsi/dr=-g<0 , this requires d rho_aug/dr<=0 . For Plummer density this is equivalent to ra/a >= sqrt(2/5)=0.6324555 (derivative at the center understood as limit). This is ONLY A NECESSARY CONDITION, not proof of DF positivity. Do not substitute the Newtonian full positivity bound 0.75 for this necessary bound. Critical obstruction to naive Step28 reuse For isolated QUMOND at large r, g_Q(r) ~ sqrt(G M a0)/r ; thus Phi(r)~sqrt(G M a0) log([excerpt]

Archived machine-readable result excerpt: {"title": "UDCT v7.14 Step29 QUMOND DF inversion preflight", "model": {"a_pc": 53.3, "stellar_mass_Msun": 32730.0, "G_pc_kms2_Msun": 0.00430091, "a0_kms2_per_pc": 3.7028130977896407, "nu": "simple", "external_field": 0}, "necessary_OM_condition": "d rho_aug / d Psi >= 0 for nonnegative f(Q), where rho_aug=(1+r\u00b2/ra\u00b2)rho; for Plummer requires ra/a >= sqrt(2/5) ~ 0.63246", "important": "Necessary condition only; passing does NOT establish positive distribution function", "obstruction": "Isolated QUMOND asymptotic potential Phi~sqrt(G*M*a0)*ln(r), diverges as r->infinity; finite-escape relative-potential Eddington inversion from Step28 cannot be reused unchanged", "samples": [{"ra_over_a": 0.5, "min_drho_aug_dPsi": -0.043405365802101, "negative_grid_points": 694, "necessary_condition_pass": false, "sign_change_r_over_a": 0.5}, {"ra_over_a": 0.6, "min_drho_aug_dPsi": -0.006271398641 [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 30 — archived source and numerical evidence

Files (15): README.md, df_positivity_scan.csv, df_q_0p5.csv, df_q_0p6.csv, df_q_0p65.csv, df_q_0p7.csv, df_q_0p75.csv, df_q_1p0.csv, df_q_inf.csv, newtonian_control.csv, newtonian_control.py, requirements.txt, results.json, spot_checks.csv, step30.py

Archived README synopsis: UDCT v7.14 Step 30 — Isolated spherical QUMOND DF inversion (exploratory) This package is an actual numerical run, not preregistration. Execute python step30.py and python newtonian_control.py (NumPy/SciPy). All data in the package are generated, not observational Hydrus I velocities. Assumptions - Plummer mass/tracer density, a=53.3 pc, stellar mass M=32730 solar masses, G=4.30091e-3 pc (km/s)^2/Msun, a0=1.2e-10 m/s². - Isolated spherical QUMOND with simple nu(y)=0.5+sqrt(0.25+1/y), g_Q=nu(g_N/a0)g_N. - Osipkov–Merritt augmented density rho_aug=(1+r²/ra²)rho(r); isotropic ra=infinity. - Positive-energy convention E=v²/2+Phi(r), Phi(0)=0, Phi(infinity)=infinity. For the unbounded potential, rho_aug(Phi) tends to zero as Phi tends to infinity. Method Abel inversion f(E) = [1/(sqrt(8) pi²)] integral_E^infinity rho_aug''(Phi)/sqrt(Phi-E) dPhi. Integrate over radius after substitution ln(r/r_E)=t² to eliminate endpoint square-root singularity. rho_aug''(Phi)=[rho_aug''(r) g(r)-rho_aug'(r) g'(r)]/g(r)^3. Phi(r) integrated on 16001 log-radius points from r/a=1e-7 to 1e8 and interpolated monotonically. Results and interpretation - df_positivity_scan.csv : 11 anisotropy conditions x 43 energies; negative f values appear in all 11 scans, including isotropic case. These are findings for this specific model , not general no-go statements for MOND, QUMOND, or UDCT. - df_q_ .csv : selected full DF curves. - newtonian_control.csv : numerica [excerpt]

Archived machine-readable result excerpt: {"model": {"a_pc": 53.3, "M_Msun": 32730.0, "a0_kms2_per_pc": 3.7028130977896407, "nu": "simple", "external_field": 0}, "inversion": "unbounded-potential Abel inversion, E=v\u00b2/2+Phi, Phi(0)=0; f(E)=1/(sqrt(8) pi\u00b2) integral_E^infinity d\u00b2rho_aug/dPhi\u00b2 /sqrt(Phi-E) dPhi", "samples": [{"ra_over_a": 0.5, "n_energy": 43, "min_df": -0.0019750387878247374, "max_df": 0.002773173822334719, "min_df_over_maxabs": -0.7121943716322707, "negative_energy_points": 10, "first_negative_r_over_a": 0.07815010964380367, "grid_nonnegative": false}, {"ra_over_a": 0.6, "n_energy": 43, "min_df": -0.000985167071764822, "max_df": 0.000705605574052948, "min_df_over_maxabs": -1.0,

"negative_energy_points": 17, "first_negative_r_over_a": 0.014999999999999998, "grid_nonnegative": false}, {"ra_over_a": 0.6324555320336759, "n_energy": 43, "min_df": -0.0008044136936525595, "max_df": 0.000675872601766625 [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 31 — archived source and numerical evidence

Files (7): README.md, density_reconstruction.csv, requirements.txt, results.json, step30_reference.py, step31.py, tail_and_tolerance.csv

Archived README synopsis: UDCT v7.14 Step 31 — QUMOND DF sign and forward reconstruction audit Exploratory, not preregistration. Actual numerical calculations. Source is the locally preserved Step 30 step30.py , included here as step30_reference.py for portable reproduction. Execute python step31.py with NumPy/SciPy. Outputs tail_and_tolerance.csv , density_reconstruction.csv , and results.json . Test - Isolated spherical QUMOND, simple nu, Plummer stellar source and tracer, a=53.3 pc, M=32730 Msun. - Isotropic and Osipkov–Merritt r_a=a. - Compare unbounded-potential DF inversion integrals terminated at r/a=1e4, 1e6, 1e8 and stricter quadrature tolerance. - Forward density reconstruction rho_aug(Phi) = 4 pi sqrt(2) integral_Phi^infty f(E) sqrt(E-Phi) dE using 145 sampled energy points up to r/a=1e6. Findings - At energy corresponding to r/a=0.025, 0.05, 0.1, DF remains negative for both q=inf and q=1. Tail and quadrature sensitivity for those values is less than ~2e-7 relative. - Forward density reconstruction near center (r/a <=0.2) agrees to ~0.08% or better, but error is several percent at r/a=1 to 3 and ~5-11% at r/a=10. Therefore full reconstruction convergence is NOT established. - One integration warning was emitted by scipy for one sampled integral; retain it in the audit trail. - These findings concern a particular imposed tracer profile and potential; do not infer QUMOND/MOND/UDCT general failure, 3D EFE behavior, or real Hydrus I fit. Reprod [excerpt]

Archived machine-readable result excerpt: {"source_step30": "/mnt/data/UDCT_v7_14_Step30_QUMOND_DF_Inversion/step30.py", "audit": "tail 1e4/1e6/1e8 and quadrature tolerance; forward density reconstruction using sampled DF", "tail_and_tolerance": [{"q": "inf", "r_over_a": 0.025, "E": 0.06056610798725432, "f_tail_1e4": -0.004668859500130536, "f_tail_1e6": -0.004668859487051844, "f_tail_1e8": -0.004668858612572882, "f_tight": -0.004668859470876699, "rel_tail_change": 1.8730037358126404e-07, "rel_tolerance_change": 1.8383586446806823e-07}, {"q": "inf", "r_over_a": 0.05, "E": 0.17168961758065898, "f_tail_1e4": -0.0026730157462840818, "f_tail_1e6": -0.0026730151466690705, "f_tail_1e8": -0.002673015209444798, "f_tight": -0.0026730157362927997, "rel_tail_change": 2.3484986955145043e-08, "rel_tolerance_change": 1.9709872811583676e-07}, {"q": "inf", "r_over_a": 0.1, "E": 0.486376410016744, "f_tail_1e4": -0.0009156494354252036, "f_tail_1e6 [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 32 — archived source and numerical evidence

Files (6): README.md, density_convergence.csv, requirements.txt, results.json, step30_reference.py, step32.py

Archived README synopsis: UDCT v7.14 Step 32 — exploratory DF reconstruction convergence This is a pre-preregistration numerical audit , not a claim of observational validation or a general no-go theorem. Model Isolated spherical QUMOND simple interpolation, Plummer stellar source and tracer, scale radius 53.3 pc, stellar mass 5 × 6546 solar masses. Compare isotropic and Osipkov–Merritt anisotropy radius r_a=a. The gravitational potential increases logarithmically at large radius; energy convention E=v²/2+Phi, Phi(0)=0. Method Run python step32.py (requires numpy and scipy). It evaluates the Abel-inverted DF at 145, 289, and 577 logarithmically spaced radii from 0.01a to 1e8a and forward-integrates the sampled DF against sqrt(E-Phi) to reconstruct the augmented tracer density. step30_reference.py preserves original Step30 formulas. Interpretation The density reconstruction errors decrease as energy grid resolution increases. Negative DF samples persist. This is an exploratory self-consistency check, not independent confirmation , because forward and inverse calculations share the same potential and derivatives. No complete 3D external field, Hydrus I star-by-star likelihood, or original v7.13 24-candidate sign test is performed. Known limitations Finite energy range, finite radial extent, numerical quadrature, sparse outer-grid energy samples; positivity cannot be certified globally. The DF sign must be tested with an independent inversion and distribu [excerpt]

Archived machine-readable result excerpt: {"method": "QUMOND OM DF inversion; forward density trapezoid over energy samples, Emax corresponds to r/a=1e8", "rows": [{"q": "isotropic", "N": 145, "r_over_a": 0.05, "reconstructed": 0.9948626326410831, "target": 0.9937772415622736, "relative_error": 0.00109218749777682, "negative_df_grid": 17}, {"q": "isotropic", "N": 145, "r_over_a": 0.1, "reconstructed": 0.9764657788798482, "target": 0.9754310265758153, "relative_error": 0.0010608154506478696, "negative_df_grid": 17}, {"q": "isotropic", "N": 145, "r_over_a": 0.2, "reconstructed": 0.9070879265569328, "target":

0.906601956075185, "relative_error": 0.0005360351127540571, "negative_df_grid": 17}, {"q": "isotropic", "N": 145, "r_over_a": 0.5, "reconstructed": 0.5682548800577408, "target": 0.5724334022399462, "relative_error": -0.00729957784757968, "negative_df_grid": 17}, {"q": "isotropic", "N": 145, "r_over_a": 1.0, "reconstructed": 0. [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 33 — archived source and numerical evidence

Files (6): README.md, central_sign_check.csv, requirements.txt, results.json, step30_reference.py, step33.py

Archived README synopsis: UDCT v7.14 Step 33 — Independent Central Asymptotic Sign Audit Status: Exploratory, pre-preregistration. This is NOT a direct Hydrus I stellar-velocity fit and NOT a general no-go theorem for QUMOND. Scope Isolated spherical simple-QUMOND gravitational field generated by a Plummer stellar mass profile. The tracer is the same Plummer profile, assumed isotropic and in stationary equilibrium. No external field, no 3D anisotropic equilibrium. Independent analytic check For r << a, g_N ~ (G M/a^3) r and simple-QUMOND g ~ sqrt(a0 G M/a^3) sqrt(r). Hence Phi(r)-Phi(0) ~ K r^(3/2), K=(2/3)sqrt(a0 G M/a^3). Plummer rho/rho0 ~ 1 - (5/2)(r/a)^2, giving rho/rho0 ~ 1-B Phi^(4/3), B=5/(2 a² K^(4/3)). The unbounded-potential isotropic Abel inversion is f(E)=1/(sqrt(8) pi²) integral_E^infinity rho''(Phi)/sqrt(Phi-E) dPhi. Since rho'' ~ -(4/9)B Phi^(-2/3), the central contribution is f(E) ~ -4 B Beta(1/2,1/6)/(9 sqrt(8) pi²) E^(-1/6), E->0+. The coefficient is strictly negative, provided the stated model assumptions hold. The outer integral contributes less singular terms. Run python step33.py (requires numpy and scipy). Produces central_sign_check.csv and results.json. step30_reference.py is the archived numerical inversion implementation; step33.py uses its functions only for the numerical side of the comparison. The analytic argument above does not use that numerical inversion. Limitations - The numerical values do not yet agree quantitativ [excerpt]

Archived machine-readable result excerpt: {"model": "isolated spherical simple-QUMOND; Plummer self-consistent stellar tracer; no external field", "analytic_derivation": {"g_small_r": "sqrt(a0 G M/a^3) r^(1/2)", "Phi_small_r": "K r^(3/2)", "rho_small_Phi": "rho0 [1 - B Phi^(4/3)]", "d2rho_dPhi2": "-(4/9) B Phi^(-2/3)", "df_small_E": "-(4 B/(9 sqrt(8) pi^2)) Beta(1/2,1/6) E^(-1/6)", "K": 0.039114429749681, "B": 0.06627802928022407, "leading_coefficient": -0.0076882700941727005}, "rows": [{"r_over_a": 0.0002, "E_kms2": 4.307594079717392e-05, "numerical_f": -0.03339335430797165, "asymptotic_f": -0.04106351956177551, "ratio_numeric_to_asymptotic": 0.8132121811364721, "sign_agrees": true}, {"r_over_a": 0.0004, "E_kms2": 0.00012186900121342507, "numerical_f": -0.026868234315503583, "asymptotic_f": -0.034528662564260876, "ratio_numeric_to_asymptotic": 0.7781429201174368, "sign_agrees": true}, {"r_over_a": 0.0008, "E_kms2": 0.0003448223 [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 34 — archived source and numerical evidence

Files (6): README.md, central_om_comparison.csv, requirements.txt, results.json, step30_reference.py, step34.py

Archived README synopsis: UDCT v7.14 exploratory Step 34 — central OM DF sign audit This is pre-preregistration exploratory work , not a UDCT validation or a direct fit to Hydrus I stars. Model Isolated spherical Plummer stellar mass and tracer, M=32730 Msun, a=53.3 pc; simple-QUMOND acceleration g = gN/2 + sqrt(gN²/4 + a0 gN) , a0=1.2e-10 m/s². OM augmented density rho_aug=(1+r²/ra²) rho_Plummer . Potential defined with Phi(0)=0 and diverges logarithmically at infinity. Numerical inversion core is the archived Step30 code, copied unchanged. Local derivation As r->0, g ~ sqrt(a0 GM/a³) r^(1/2), so Phi ~ K r^(3/2), K=(2/3)sqrt(a0 GM/a³). Normalize rho0=1. With q=ra/a, rho_aug=1+(q^-2-5/2)(r/a)²+O(r^4). Thus rho_aug(Phi)=1+D Phi^(4/3)+... where D=(q^-2-5/2)/(a² K^(4/3)). The unbounded-potential Abel inversion yields f(Q)~ [4 Beta(1/2,1/6)/(9 sqrt(8) pi²)] D Q^(-1/6). The sign of the leading central term flips at q=sqrt(2/5). At equality the leading term vanishes and higher-order terms are necessary. Run pip install -r requirements.txt python step34.py Outputs: central_om_comparison.csv , results.json . Interpretation and limits 54 direct numerical comparisons across 9 anisotropy choices and 6 inner radii. Numerical comparison uses the same Step30 inversion implementation: analytic local derivation is independent, but there is not a second independent numerical inversion algorithm. Positive central leading coefficient does NOT establish global DF positivi [excerpt]

Archived machine-readable result excerpt: {"model": "isolated spherical simple-QUMOND with Plummer stellar tracer and source", "units": "pc, km/s, Msun; normalized tracer density", "central_exponent": "f(E) proportional to E^(-1/6)", "critical_q": 0.6324555320336759, "leading_coefficient_prefactor": 0.0030753080376690806, "number_of_numerical_checks": 54, "checks": [{"q": 0.5, "r_over_a": 1e-05, "E_kms2": 4.808983807262545e-07, "numerical_f": 0.046473564660154246, "asymptotic_f": 0.05211599054099163, "ratio": 0.8917333082943256, "sign_agreement": true}, {"q": 0.5, "r_over_a": 3e-05, "E_kms2":

2.501092184387727e-06, "numerical_f": 0.0339632196650216, "asymptotic_f": 0.03959359464640946, "ratio": 0.857795811881444, "sign_agreement": true}, {"q": 0.5, "r_over_a": 0.0001, "E_kms2": 1.5226597479463714e-05, "numerical_f": 0.02367828880774576, "asymptotic_f": 0.0293007950378758, "ratio": 0.8081107962134788, "sign_agreement": true}, {"q [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 35 — archived source and numerical evidence

Files (5): README.md, independent_gauss_df.csv, requirements.txt, results.json, step35.py

Archived README synopsis: UDCT v7.14 Step 35 — Independent quadrature audit of OM DF signs Status: exploratory, pre-preregistration. Not a UDCT-specific prediction, 3D external-field test, or direct Hydrus I fit. Scope Independent numerical integration method for the unbounded-potential Abel inversion of a spherical, isolated, simple-QUMOND Plummer tracer and source. Stellar mass M=32730 Msun, Plummer a=53.3 pc, a0=1.2e-10 m/s², zero external field. The OM augmented tracer density is rho_aug=(1+r²/ra²)rho_Plummer. The independent step35.py uses fixed-order Gauss-Legendre quadrature in t=sqrt(log(r/r_E)), and direct adaptive quadrature for the potential difference Phi(r)-Phi(r_E). It does not import or call the Step30 df() or its interpolated potential. Both approaches share the analytic QUMOND acceleration and tracer derivatives; this is an independent integration algorithm, not independent physics. Results - 7 OM/anisotropy cases x 11 turning radii = 77 inversion evaluations, each with 96- and 192-point Gauss quadrature. - Maximum relative discrepancy (96 vs 192 nodes): about 0.0137%. - All seven cases contain sampled negative DF values. q=0.5 and 0.6 are positive in the central sample yet negative at intermediate turning radii. - At q=sqrt(2/5), the local x² term in rho_aug cancels, but the full inversion is numerically negative at the sampled center, ~-0.0006346. The local x^4 term is -15/8 x^4. This local coefficient does NOT determine the sign of [excerpt]

Archived machine-readable result excerpt: {"method": "independent Gauss-Legendre quadrature with direct quadrature of potential difference; does not call step30 inversion or its PCHIP potential", "model": {"a_pc": 53.3, "M_Msun": 32730.0, "a0_kms2_per_pc": 3.7028130977896407, "external_field": 0}, "q_critical": 0.6324555320336759, "rho_aug_x4_at_critical": -1.875, "n_evaluations": 77, "max_relative_96_192": 0.00013688581895453822, "critical_rows": [{"q": 0.6324555320336759, "r_over_a": 1e-05, "df_gauss96": -0.0006345193475164711, "df_gauss192": -0.0006346062161080767, "relative_delta": 0.00013688581895453822, "sign": "negative"}, {"q": 0.6324555320336759, "r_over_a": 0.0001, "df_gauss96": -0.0006346265930858828, "df_gauss192": -0.0006346159355512559, "relative_delta": 1.6793676348026132e-05, "sign": "negative"}, {"q": 0.6324555320336759, "r_over_a": 0.001, "df_gauss96": -0.000634891956121354, "df_gauss192": -0.000634892402140615 [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 36 — archived source and numerical evidence

Files (5): README.md, requirements.txt, results.json, step36.py, tracer_scale_scan.csv

Archived README synopsis: UDCT v7.14 Step 36 — tracer-scale sensitivity (exploratory) Run: python step36.py (requires numpy and scipy). Output: tracer_scale_scan.csv and results.json. Fixed baryonic Plummer mass M=32730 Msun and scale a=53.3 pc; isotropic Plummer tracer scale b/a = 0.5,0.75,1,1.5,2. The tracer is normalized to unit central density, so reported DF amplitudes are dimensionless-normalized and only sign comparisons are intended. Center-referenced potential, isolated spherical simple-QUMOND g = (gN+sqrt(gN^2+4 a0 gN))/2. Integration uses r=r0 exp(t²) and direct potential quadrature. All 25 tests are exploratory, not preregistered. The sampled central energies yield negative DF for all tracer scales, but there is no all-energy proof, no external field, no 3D equilibrium, no individual-star Hydrus I fit, and no UDCT-specific physics. A negative isotropic DF for this specified tracer/potential pair does not falsify QUMOND generally. Step 35 and earlier data are not embedded; this script is self-contained.

Archived machine-readable result excerpt: {"cases": 25, "negative": 19, "positive": 6, "model": "Isolated spherical simple QUMOND, fixed Plummer gravitating mass profile and variable Plummer isotropic tracer scale", "a_pc": 53.3, "M_msun": 32730, "limitations": ["25 sampled energies only", "No external field", "Not a full 3D Jeans fit", "Not a test of UDCT-specific terms", "No Hydrus I individual-star likelihood"]}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 37 — archived source and numerical evidence

Files (6): README.md, cusp_profile_scan.csv, requirements.txt, results.json, run.log, step37.py

Archived README synopsis: UDCT v7.14 Step 37 — Central tracer-shape sensitivity (exploratory) Run python step37.py (Python 3, NumPy, SciPy). Outputs cusp_profile_scan.csv and results.json . Setup Fixed baryonic Plummer source: M=32730 Msun, a=53.3 pc, isolated spherical simple-QUMOND acceleration. Normalized stellar tracer density: rho(r)/rho(0)=[1+(r/a)^p]^(-5/p). The tracer profile is finite at r=0 for all p. Here 'central shape' is more accurate than 'cusp': p<2 has a nonanalytic center, but density does not diverge. The distribution function is computed via a center-referenced, unbounded-potential isotropic Abel inversion. p=2 recovers Plummer tracer. This is a controlled model-family sensitivity, not a realistic profile fit. Findings 40 evaluations (8 p values x 5 energy turning radii). p=1,1.25,1.5 are positive at all sampled points; p>=1.75 has at least one negative sampled value. The sign at small energies depends strongly on the tracer center-shape assumption. This does not imply all-energy positivity for p<=1.5. Limits No external field; no anisotropic 3D equilibrium; no full Hydrus I individual-star fit; no claim about UDCT-specific gravity or QUMOND generally. Values normalized to unit central tracer density; compare signs, not amplitudes across p.

Archived machine-readable result excerpt: {"step": 37, "n_cases": 40, "negative": 20, "positive": 20, "p_values": [1.0, 1.25, 1.5, 1.75, 2.0, 2.5, 3.0, 4.0], "r_over_a": [0.0001, 0.001, 0.01, 0.1, 0.5], "model": "fixed isolated spherical simple-QUMOND gravity from Plummer baryonic mass; tracer rho=(1+(r/a)^p)^(-5/p)", "note": "The density family has a finite central value for all p, with different approach-to-center exponents; not a divergent density cusp.", "cautions": ["Sampled energies only, not global DF positivity", "No external field or 3D DF", "No direct Hydrus I observational fit", "Numerical quadrature error is not a physical systematic error bound"]}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 38 — archived source and numerical evidence

Files (5): UDCT_v7_14_Step38_Full_Energy_Audit/README.md, UDCT_v7_14_Step38_Full_Energy_Audit/full_energy_scan.csv, UDCT_v7_14_Step38_Full_Energy_Audit/requirements.txt, UDCT_v7_14_Step38_Full_Energy_Audit/results.json, UDCT_v7_14_Step38_Full_Energy_Audit/step38.py

Archived README synopsis: UDCT v7.14 — Step 38: expanded energy and central-shape audit Status: exploratory, not preregistration, not an observational fit. Model - Spherical isolated simple-QUMOND acceleration from Plummer baryonic mass M=32730 solar masses, a=53.3 pc, a0=1.2e-10 m/s². - Isotropic tracer density normalized to unity at center: rho=(1+(r/a)^p)^(-5/p). - Center-referenced energy inversion in an unbounded logarithmic asymptotic potential; the tabulated r is the energy-turning radius, not an observed star position. - Log-radius potential integration and PCHIP interpolation; Gauss–Legendre integration with 256 and 512 nodes for a numerical comparison. Reproduce python step38.py requires NumPy and SciPy. Regenerates full_energy_scan.csv and results.json . Result 13 p values times 91 log-spaced turning radii in [1e-6 a, 1e3 a]. All 91 sampled values nonnegative for p<=1.5; negative values appear at p=1.55 and higher. The observed onset is bounded by the tested grid (1.50, 1.55], not a proven physical critical exponent. Numerical limitations The difference between 256- and 512-node estimates is a quadrature convergence check only. Both use the same interpolated potential and derivative formulas, so systematic bias is shared. For p=1 the largest absolute N-difference is ~1.96e-6; this deserves targeted refinement. Potential interpolation starts at r/a=1e-10, not r=0; finite outer radius r/a=1e9 for inversion; far-tail and center asymptotics are [excerpt]

Archived machine-readable result excerpt: {"model": "fixed Plummer gravity, isotropic generalized cored tracer, isolated simple QUMOND", "sampled_r_over_a": [1e-06, 1000.0], "n_per_p": 91, "n": 1183, "by_p": {"1.0": {"negative_count": 0, "first_negative_r_over_a": null, "last_negative_r_over_a": null, "max_abs_N_difference": 1.9563522073440254e-06}, "1.25": {"negative_count": 0, "first_negative_r_over_a": null, "last_negative_r_over_a": null, "max_abs_N_difference": 3.0942146622692235e-08}, "1.4": {"negative_count": 0, "first_negative_r_over_a": null, "last_negative_r_over_a": null, "max_abs_N_difference": 1.5587602320010774e-09}, "1.5": {"negative_count": 0, "first_negative_r_over_a": null, "last_negative_r_over_a": null, "max_abs_N_difference": 1.1868978022633314e-13}, "1.55": {"negative_count": 39, "first_negative_r_over_a": 1e-06, "last_negative_r_over_a": 0.006309573444801936, "max_abs_N_difference": 9.805889433778248e-11}, [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 39 — archived source and numerical evidence

Files (5): UDCT_v7_14_Step39_Critical_Exponent/README.md, UDCT_v7_14_Step39_Critical_Exponent/critical_scan.csv, UDCT_v7_14_Step39_Critical_Exponent/requirements.txt, UDCT_v7_14_Step39_Critical_Exponent/results.json, UDCT_v7_14_Step39_Critical_Exponent/step39.py

Archived README synopsis: UDCT v7.14 — Step 39: Critical exponent audit (exploratory) Scope Follow-up to Step 38. Reuses its explicit spherical simple-QUMOND acceleration prescription and generalized cored tracer density, but runs a new focused 81-point numerical grid. This is not a fit to individual Hydrus I velocities, not a 3D external-field calculation, and not a preregistered result. Model - Fixed Plummer gravitating mass: M=32730 solar masses, a=53.3 pc; G=4.30091e-3 pc (km/s)^2 / solar mass. - Isolated spherical simple-QUMOND acceleration: g=(g_N/2)+sqrt(g_N^2/4+a0 g_N), a0=1.2e-10 m/s^2. - Isotropic tracer density rho=(1+(r/a)^p)^(-5/p) (normalization rho0=1). - Potential increases outward, Psi(r)=Phi(r)-Phi(0). Energy E=Psi(r0). The inversion is the unbounded-potential Abel inversion f(E)=1/(sqrt(8) pi^2) integral_E^infinity rho''(Psi)/sqrt(Psi-E) dPsi, with boundary assumptions implicit in the formula. The integral is transformed to log radius and regularized near the lower endpoint. Asymptotic prediction At small r: g_N~r, g~r^(1/2), Psi~r^(3/2). The tracer behaves as rho~rho0-A Psi^(2p/3). For p != 1.5, the leading singular contribution to f(E) is proportional to (1-2p/3) E^(2p/3-3/2) with positive prefactor in the applicable asymptotic regime (p near 1.5). Thus p=1.5 is a central asymptotic sign boundary. At p=1.5 the leading coefficient cancels and subleading/nonlocal terms must be retained. Numerical verification python step39.py writes [excerpt]

Archived machine-readable result excerpt: {"model": "isolated spherical simple-QUMOND fixed Plummer gravity; isotropic generalized cored tracer", "critical_asymptotic_p": 1.5, "sample_count": 81, "radius_samples": [1e-06, 1e-05, 0.0001, 0.001, 0.01, 0.1, 1, 10, 100], "p_samples": [1.45, 1.49, 1.499, 1.5, 1.501, 1.505, 1.51, 1.525, 1.55], "counts": {"1.45": 0, "1.49": 0, "1.499": 0, "1.5": 0, "1.501": 2, "1.505": 3, "1.51": 3, "1.525": 4, "1.55": 4}, "caveats": ["finite radius grid does not establish global positivity", "the p=1.5 leading singular coefficient vanishes and subleading terms matter", "quadrature convergence does not validate physical assumptions"]}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 40 — archived source and numerical evidence

Files (5): UDCT_v7_14_Step40_Asymptotic_Audit/README.md, UDCT_v7_14_Step40_Asymptotic_Audit/audit.csv, UDCT_v7_14_Step40_Asymptotic_Audit/requirements.txt, UDCT_v7_14_Step40_Asymptotic_Audit/step40.py, UDCT_v7_14_Step40_Asymptotic_Audit/summary.json

Archived README synopsis: UDCT v7.14 — Step 40: Critical-exponent asymptotic audit Status: exploratory, NOT preregistered, not a Hydrus I star-by-star fit, not 3D EFE. Reproduction python -m pip install -r requirements.txt python step40.py This produces audit.csv (30 samples) and summary.json . The numerical calculation reuses the physical model and inversion equation of Step 39 but recomputes the potential on a denser, wider grid and evaluates 192- and 384-node Gauss–Legendre integrals. Thus it is an independently rerun numerical calculation, not an independent proof of the inversion theorem . The small differences between node counts assess only quadrature convergence. Model G=4.30091e-3 pc (km/s)^2/Msun; M=32730 Msun; a=53.3 pc; a0=1.2e-10 m/s². Plummer gravitating mass with isolated spherical simple-QUMOND acceleration gN(r)=GM r/(r²+a²)^(3/2) g(r)=gN/2 + sqrt(gN²/4+a0 gN) Tracer density rho_p(r)=[1+(r/a)^p]^(-5/p), with rho(0)=1. Potential Psi=Phi(r)-Phi(0), E=Psi(r0), and the unbounded-potential isotropic Abel inversion f(E) = [1/(sqrt(8) pi²)] integral_E^infinity rho''(Psi)/sqrt(Psi-E) dPsi. The inversion assumes adequate decay at infinity and a stationary isotropic DF f(v²/2+Psi); it does not establish physical stability or uniqueness beyond these assumptions. New asymptotic calculation Let k=GM/a³ and h=sqrt(a0 k). As r→0: g(r)=h r^(1/2)+(k/2)r+O(r^(3/2)) Psi(r)=A r^(3/2)+(k/4)r²+O(r^(5/2)), A=2h/3. For p near 3/2, set alpha=2p/3. Then rho(Psi [excerpt]

Archived machine-readable result excerpt: {"k": 0.0009296612580853805, "h": 0.0586716446245215, "A": 0.039114429749681, "B": 0.21900343602734065, "Dcoef": 0.003833537061489486, "critical_coefficient": 0.00044469138061029365, "results": [{"p": 1.49, "r_over_a": 0.0001, "E": 1.5227065691232654e-05, "f192": 0.055313492447964636, "f384": 0.05531349244775056, "relative_difference": 3.8702809251389406e-12, "leading_asymptotic": 0.04574810906563048, "critical_subleading": null}, {"p": 1.49, "r_over_a": 1e-05, "E": 4.813792867894178e-07, "f192": 0.27525408859922806, "f384": 0.27525408860028966, "relative_difference": 3.856782878484943e-12, "leading_asymptotic": 0.2632924572664668, "critical_subleading": null}, {"p": 1.49, "r_over_a": 1e-06, "E": 1.5221121830168048e-08, "f192": 1.531483468575598, "f384": 1.5314834685813654, "relative_difference": 3.76588235305287e-12, "leading_asymptotic": 1.5151619584003162, "critical_subleading": null} [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 41 — archived source and numerical evidence

Files (6): README.md, comparison.csv, independent_adaptive.csv, requirements.txt, step41.py, summary.json

Archived README synopsis: UDCT v7.14 — Step 41: Independent quadrature audit Status: pre-preregistration exploratory numerical audit, 2026-10-25. What was independently changed Step 40 used a tabulated/interpolated potential and Gauss–Legendre quadrature. This step computes potential differences by adaptive scipy.integrate.quad on every integrand evaluation, and independently evaluates the outer inversion integral using adaptive QUAD. No potential interpolation or Legendre nodes are used. The same underlying isotropic Abel inversion equation and the same spherical Plummer-QUMOND physical assumptions are retained; this is NOT an independent derivation of that equation. Model G=4.30091e-3 pc (km/s)^2/Msun; M=32730 Msun; a=53.3 pc; a0=1.2e-10 m/s^2. gN=G M r/(r r+a a)^1.5; g=gN/2+sqrt(gN gN/4+a0 gN). Tracer rho=(1+(r/a)^p)^(-5/p), isotropic DF; potential Phi(r)-Phi(0) unbounded as r tends to infinity. f(E) = 1/(sqrt(8) pi^2) integral_E^infty rho''(Psi)/sqrt(Psi-E) dPsi. The integral is numerically truncated at r=1e6 a, with the tail not rigorously bounded in this step. Run pip install -r requirements.txt python step41.py Outputs: independent_adaptive.csv, summary.json. comparison.csv compares against Step 40 (precomputed, included in ZIP). 21 calculations: 7 exponents by 3 radii. The calculations agree closely with Step 40, supporting numerical implementation robustness rather than independent physical validation. Cautions A local center asymptotic sign b [excerpt]

Archived machine-readable result excerpt: {"method": "adaptive scipy QUAD with potential integrated on demand; no PCHIP; independent quadrature, same analytical inversion identity", "rows": [{"p": 1.49, "r_over_a": 0.0001, "f_adaptive": 0.05531349280616654, "quad_error_est": 2.6648634534324764e-12, "sign": "positive"}, {"p": 1.49, "r_over_a": 1e-06, "f_adaptive": 1.531483478664897, "quad_error_est": 5.325634573572877e-09, "sign": "positive"}, {"p": 1.49, "r_over_a": 1e-08, "f_adaptive": 50.211413622180125, "quad_error_est": 6.187731496136262e-08, "sign": "positive"}, {"p": 1.499, "r_over_a": 0.0001, "f_adaptive": 0.013553625041341335, "quad_error_est": 2.1112265267460742e-11, "sign": "positive"}, {"p": 1.499, "r_over_a": 1e-06, "f_adaptive": 0.15038599373729092, "quad_error_est": 4.780794570839916e-09, "sign": "positive"}, {"p": 1.499, "r_over_a": 1e-08, "f_adaptive": 4.321045926663445, "quad_error_est": 4.148130793805234e-08, " [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 42 — archived source and numerical evidence

Files (5): README.md, density_reconstruction.csv, requirements.txt, step42.py, summary.json

Archived README synopsis: UDCT v7.14 Step 42 — Forward density reconstruction audit Status: exploratory, pre-preregistration. 2026-10-25. Aim Test whether the signed distribution function recovered using the right-sided Abel inversion reconstructs the input tracer density. This is an internal consistency check, not a proof of physical DF positivity, observational fit, or a general failure of QUMOND. Model Spherical isolated QUMOND with Plummer gravitating mass M=32730 Msun, a=53.3 pc, acceleration scale a0=1.2e-10 m/s^2, simple nu(x)=0.5+sqrt(0.25+1/x). Tracer density rho(r)/rho0=[1+(r/a)^p]^(-5/p). Spherical potential is defined by dPhi/dr=g(r)>0; its zero-point cancels from integrals. In an isolated deep-MOND exterior, Phi diverges logarithmically, so the relevant inversion is the right-sided Abel transform over Phi to +infinity. rho(Phi)=4 pi sqrt(2) integral_Phi^infinity f(E) sqrt(E-Phi) dE f(E)=1/(sqrt(8) pi^2) integral_E^infinity rho''(Phi)/sqrt(Phi-E) dPhi Procedure The Python script evaluates g, dg/dr, rho', rho'', constructs Phi(r) by cumulative quadrature on a logarithmic radial grid, obtains rho''(Phi), performs inverse Abel quadrature with endpoint substitution, then forward-integrates the resulting signed f to recover rho. Compare 900 and 1400 radial grid points; Gauss-Legendre order 112. Outer cutoff rmax/a=1e6. Reconstruction at r/a=1e-4,1e-3,1e-2,0.1,0.5,1,3,10. Four density exponents p=1.49,1.5,1.501,2.0. Results Maximum absolute relat [excerpt]

Archived machine-readable result excerpt: {"cases": [{"p": 1.49, "negative_samples": 0, "positive_samples": 899, "f_min": 0.0, "f_max": 0.17393836354856199, "rmax_over_a": 1000000.0, "ngrid": 900, "nq": 112}, {"p": 1.49, "negative_samples": 0, "positive_samples": 1399, "f_min": 0.0, "f_max": 0.17390295570207878, "rmax_over_a": 1000000.0, "ngrid": 1400, "nq": 112}, {"p": 1.5, "negative_samples": 0, "positive_samples": 899, "f_min": 0.0, "f_max": 0.010938369893892193, "rmax_over_a": 1000000.0, "ngrid": 900, "nq": 112}, {"p": 1.5, "negative_samples": 0, "positive_samples": 1399, "f_min": 0.0, "f_max": 0.01093727924929509, "rmax_over_a": 1000000.0, "ngrid": 1400, "nq": 112}, {"p": 1.501, "negative_samples": 340, "positive_samples": 559, "f_min": -0.003707810430893841, "f_max": 0.007327713184931298, "rmax_over_a": 1000000.0, "ngrid": 900, "nq": 112}, {"p": 1.501, "negative_samples": 529, "positive_samples": 870, "f_min": -0.003705839 [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 43 — archived source and numerical evidence

Files (6): README.md, cutoff_scan.csv, requirements.txt, step42.py, step43.py, summary.json

Archived README synopsis: UDCT v7.14 — Step 43: Outer cutoff sensitivity audit Status: pre-preregistration exploratory computation. 2026-10-25. Scope Reuses the Step 42 inversion implementation unchanged and varies the maximum integration radius. Four tracer density exponents p = 1.49, 1.5, 1.501, 2.0; four outer cutoff radii R/a = 1e4, 1e5, 1e6, 1e7; 1000 log-spaced radial grid points and 128 Gauss-Legendre nodes. Eight reconstruction radii per case, 128 reconstruction comparisons in total. Potential and density assumptions: fixed spherical Plummer baryonic gravitational source with a=53.3 pc, M=32730 Msun, G=4.30091e-3 pc (km/s)^2/Msun; spherical QUMOND algebraic acceleration g = gN/2 + sqrt(gN^2/4 + a0 gN), a0=1.2e-10 m/s^2; isotropic tracer rho=(1+(r/a)^p)^(-5/p). Run From the extracted archive directory, run: python -m pip install -r requirements.txt python step43.py step43.py imports step42.py , included here as a frozen dependency. Outputs cutoff_scan.csv and summary.json . Results At R/a=1e4, 1e5, 1e6, 1e7 respectively, negative grid sample counts: - p=1.49: 0,0,0,0 - p=1.5: 0,0,0,0 - p=1.501: 434,406,382,360 - p=2.0: 678,636,598,565 Max absolute relative density reconstruction errors across the 8 radii for each cutoff are approximately 0.0336%, 0.0272%, 0.0434%, 0.0497%. Results depend slightly on p. Important limitations 1. This is not a rigorous bound on the omitted infinite-radius integral. A bound requires asymptotic tail analysis or a con [excerpt]

Archived machine-readable result excerpt: {"cases": [{"p": 1.49, "cutoff_over_a": 10000.0, "max_abs_relative_density_error": 0.00033635015477789294, "negative_grid_samples": 0, "minimum_df": 0.0}, {"p": 1.49, "cutoff_over_a": 100000.0, "max_abs_relative_density_error": 0.0002723203119641741, "negative_grid_samples": 0, "minimum_df": 0.0}, {"p": 1.49, "cutoff_over_a": 1000000.0, "max_abs_relative_density_error": 0.00043391988965539114, "negative_grid_samples": 0, "minimum_df": 0.0}, {"p": 1.49, "cutoff_over_a": 10000000.0, "max_abs_relative_density_error": 0.0004971356511194214, "negative_grid_samples": 0, "minimum_df": 0.0}, {"p": 1.5, "cutoff_over_a": 10000.0, "max_abs_relative_density_error": 0.000336361956061555, "negative_grid_samples": 0, "minimum_df": 0.0}, {"p": 1.5, "cutoff_over_a": 100000.0, "max_abs_relative_density_error": 0.0002723346855569703, "negative_grid_samples": 0, "minimum_df": 0.0}, {"p": 1.5, "cutoff_over_a [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 44 — archived source and numerical evidence

Files (5): README.md, fixed_energy_tail.csv, requirements.txt, step44.py, summary.json

Archived README synopsis: UDCT v7.14 — Step 44: Fixed-energy outer-tail audit Status: exploratory, not preregistration , 2026-10-25. Scientific question Does the negative isotropic distribution function in a specified isolated spherical Plummer–QUMOND model depend on truncating the outer radial integration? Model and method Gravitating mass: spherical Plummer with a=53.3 pc, M=32730 solar masses. G=4.30091e-3 pc (km/s)^2 / Msun; a0=1.2e-10 m/s^2. Spherical acceleration g=gN/2+sqrt(gN^2/4+a0 gN). Isotropic tracer density rho=(1+(r/a)^p)^(-5/p). p in [1.49,1.5,1.501,2.0]. The isotropic inversion is f(E)=1/(sqrt(8) pi^2) integral_E^infty [d²rho/dPhi²] / sqrt(Phi-E) dPhi, with the boundary derivative at infinity vanishing for the chosen tracer. E=Phi(r_E). We compute Phi(r)-Phi(r_E) by solving dPhi/d(log r)=r g(r) using an adaptive ODE solver. The change of variables log(r/r_E)=t² removes the lower-end Abel square-root singularity. The radial integral is then evaluated using scipy.integrate.quad. No PCHIP interpolation of the potential or Gauss-Legendre quadrature from Step 43 is used. Cutoffs R/a = 1e4,1e5,1e6,1e7,1e9,1e12; reference R/a=1e14 (finite). For each of 4 p and 4 fixed physical energies (r_E/a=1e-4,0.01,0.1,1), 6 cutoff comparisons: 96 rows. Findings At r_E/a=1e-4, reference f values: p=1.49: +0.05531349112; p=1.5: +0.009280164943; p=1.501: +0.005076663140; p=2: -0.04115614022. At p=1.501, f(r_E/a=1e-5) ≈ -0.012001 and f(r_E/a=1e-6) ≈ -0.116173 [excerpt]

Archived machine-readable result excerpt: {"cases": 96, "reference_outer_radius_over_a": 100000000000000.0, "reference_is_finite_not_proven_infinite": true, "max_abs_delta_by_cutoff": {"10000.0": 8.303911464696556e-11, "100000.0": 6.683233827464719e-11, "1000000.0": 6.785093754207994e-11, "10000000.0": 8.324192030118027e-11, "1000000000.0": 1.1650182207834092e-10, "1000000000000.0": 1.2485647932214405e-10}, "min_abs_reference_df": 0.00046453404229980684, "warnings": ["Numerical tail estimate, not a rigorous mathematical bound", "Same spherical QUMOND field and isotropic Abel inversion assumptions", "Fixed physical energy (rE/a), unlike earlier changing-grid sample counts", "Reference 1e14 a is finite; no claim of exact infinity"]}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 45 — archived source and numerical evidence

Files (6): README.md, om_scan.csv, requirements.txt, step45.py, summary.csv, summary.json

Archived README synopsis: UDCT v7.14 — Step 45: OM Anisotropy Audit Status: pre-preregistration exploration, 2026-10-25. Objective Test whether the central negative distribution function in the isolated spherical Plummer-QUMOND field persists when the generalized cored tracer is assigned Osipkov–Merritt (OM) anisotropy. Model - Gravitating Plummer source: mass M=32730 Msun, scale a=53.3 pc. - Acceleration in spherical QUMOND: g = g_N/2 + sqrt(g_N^2/4 + a0 g_N), with a0=1.2e-10 m/s^2. - Tracer density normalized to unity at center: rho=(1+(r/a)^p)^(-5/p). - OM augmented tracer: rho_Q=(1+r^2/ra^2)rho. - OM inversion of rho_Q with respect to the potential Phi(r) with Phi'(r)=g(r); the zero of Phi is arbitrary and the potential diverges logarithmically at infinity. - Isotropic case ra=infinity. - Formula: f(Q)=(1/(sqrt(8) pi^2)) integral_{Phi(Q)}^infty [d²rho_Q/dPhi²]/sqrt(Phi-Phi(Q)) dPhi, where Q denotes the appropriate OM energy coordinate under this convention. - Numerically integrate in logarithmic radius with a square-root change of variables, solving potential differences with solve_ivp and integrating with scipy.quad. Reproduce Python 3 with numpy and scipy: pip install -r requirements.txt python step45.py Outputs om_scan.csv , summary.csv , and summary.json . Results 4 p values × 5 ra/a values × 6 radius-to-energy mappings = 120 computed points; 16 negative. See summary.csv for per-model counts and om_scan.csv for each value. Notably, at p=1.501, [excerpt]

Archived machine-readable result excerpt: {"n_cases": 120, "n_negative": 16, "model": "spherical isolated QUMOND with generalized cored tracer and OM augmented density", "outer_cutoff_a": 1000000000.0, "notes": ["OM f(Q) is not the same as isotropic f(E)", "finite grid: no claim of positivity over all energies", "no actual Hydrus I individual star velocity fit", "common potential and inversion assumptions"]}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 46 — archived source and numerical evidence

Files (6): README.md, om_2d_scan.csv, requirements.txt, sign_map_summary.csv, step46.py, summary.json

Archived README synopsis: UDCT v7.14 — Step 46: OM anisotropy sign map Status: exploratory, before preregistration . No Hydrus I individual-star likelihood fitting. Purpose Map signs of an Osipkov–Merritt (OM) distribution function in an isolated spherical QUMOND potential sourced by a Plummer mass distribution. Tracer density: rho(r) = [1+(r/a)^p]^(-5/p). OM augmented density: (1+r^2/r_a^2) rho(r). Computation python step46.py regenerates om_2d_scan.csv , sign_map_summary.csv , and summary.json . - 11 p values: 1.45, 1.48, 1.49, 1.495, 1.5, 1.501, 1.505, 1.51, 1.55, 1.6, 2.0 - 10 anisotropy radii r_a/a: 0.4, 0.5, 0.6, 0.65, 0.75, 1.0, 1.5, 2.0, 4.0, infinity (isotropic) - 33 log-spaced energy-associated turning radii from 1e-7 a to 1e2 a - 3630 f(Q) values, of which 800 are negative - 96- and 192-point Gauss–Legendre quadrature. No sign disagreements; largest absolute difference 2.87275e-5 in the chosen normalized units. Method The force g(r) and derivative dg/dr are analytic; potential differences are computed by a high-accuracy ODE integration over log radius. The OM augmented density derivatives are analytic. The Eddington/OM Abel inverse integral is computed over log radius with a square-root-endpoint-removing substitution. Outer numerical radius is 1e9 a. r_energy_over_a labels the radial point where Q=Phi(r), not an observed star radius. Interpretation and limitations - p=1.5 is not a universal positivity boundary for anisotropic models. For r_a [excerpt]

Archived machine-readable result excerpt: {"cases": 3630, "negative_cases": 800, "max_abs_quadrature_difference": 2.8727503739389704e-05, "n_p": 11, "n_ra": 10, "n_energy": 33, "caveat": "Finite sampled energies; zero negatives does not certify globally nonnegative DF; same analytic gravity and OM inversion used at both quadrature orders."}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 47 — archived source and numerical evidence

Files (7): README.md, exact_om_zero_contour.csv, refined_energy_zero_crossings.csv, requirements.txt, step46_om_2d_scan_input.csv, step47.py, summary.json

Archived README synopsis: UDCT v7.14 — Step 47: OM zero-contour audit Status: exploratory, not preregistered , 2026-10-25. Reproduction 1. Install pip install -r requirements.txt . 2. Place the original Step 46 om_2d_scan.csv at ../UDCT_v7_14_Step46_OM_Sign_Map/om_2d_scan.csv and run python step47.py . 3. The ZIP also includes a frozen copy of Step 46 input as step46_om_2d_scan_input.csv . To run from the ZIP alone, the script automatically falls back to this local copy. Mathematics For the fixed spherical gravitational potential and the Osipkov–Merritt augmented tracer density, rho_OM(r) = rho(r) [1 + (r/r_a)^2] , the DF inversion is linear. With r_a/a dimensionless, f(Q;r_a) = f_iso(Q) + B(Q)/(r_a/a)^2 . At fixed energy the finite positive zero contour is r_a/a = sqrt(-B/f_iso) when B f_iso < 0 . This identity is algebraic, not an independent physical proof of DF admissibility. Results - 3,630 Step 46 input rows; 363 distinct (p, energy-radius) combinations. - Maximum absolute residual of OM affine relation across all original rows: 5.68434e-14. - 139 of 363 combinations have a finite positive zero-contour value of r_a/a (not all within tested anisotropy grid). - 13 energy-radius zero crossings refined by Brent root search using new 192-node quadrature; each independently checked with 96 and 384 nodes. - Maximum absolute 384-node DF at the 192-node roots: 3.46e-9; near steep central transitions, small root errors can magnify the DF residual. Files - [excerpt]

Archived machine-readable result excerpt: {"step46_input_rows": 3630, "affine_identity_max_absolute_error": 5.684341886080802e-14, "affine_identity_median_absolute_error": 1.734723475976807e-17, "energy_p_pairs": 363, "pairs_with_finite_ra_zero": 139, "refined_roots": 13, "root_max_abs_f384": 3.4560662732693745e-09, "root_max_abs_f384_minus_f192": 3.4560726653861935e-09}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 48 — archived source and numerical evidence

Files (6): README.md, requirements.txt, roots_and_minimum.csv, step47_reference.py, step48.py, summary.json

Archived README synopsis: UDCT v7.14 Step 48 — negative OM distribution-function lobe Exploratory audit, not preregistration. Model: fixed isolated spherical Plummer-generated QUMOND-like acceleration, tracer density rho=(1+(r/a)^p)^(-5/p), OM augmented density rho_OM=(1+r^2/r_a^2)rho. Parameters p=1.5 and r_a/a=0.4. Reproduction Install pip install -r requirements.txt ; run python step48.py . step47_reference.py is bundled for standalone operation. The script computes the DF using the radial integral and a Gauss–Legendre quadrature with a numerical potential difference, searches for the two zero crossings and minimizes the DF between them, and repeats at 96, 192, 384 nodes. Findings See summary.json and roots_and_minimum.csv . The zero crossings in the energy-label radius r/a are 0.2237024398 and 0.7669697146; minimum f=-0.0008538448012 at r/a=0.3804964036. The negative-lobe width in this radius coordinate is 0.5432672748. This is not a physical radial interval containing negative stellar densities: r labels energy, and f<0 denotes failure of the assumed OM DF positivity at those energies. Caveats The 96/192/384 comparisons share the same inversion and gravity assumptions. This does not establish a global positive DF elsewhere, nor rule out other tracer or anisotropy models. No direct Hydrus I individual-star velocity fitting, 3D EFE, or v7.13 24-candidate retest is included. It is not a universal MOND failure theorem. Future work: systematic grid of [excerpt]

Archived machine-readable result excerpt: {"model": "isolated spherical Plummer-generated QUMOND-like law; p=1.5, ra/a=0.4", "left_root_r_over_a": 0.22370243982922897, "right_root_r_over_a": 0.7669697146174498, "negative_interval_width_r_over_a": 0.5432672747882208, "negative_interval_width_log_r": 1.2321305399438338, "minimum_r_over_a": 0.38049640362537734, "minimum_f_192": -0.0008538448012277041, "min_f_96": -0.0008538448011829086, "min_f_384": -0.0008538448012095206, "max_abs_384_minus_192_at_landmarks": 3.967945680355617e-14}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 49 — archived source and numerical evidence

Files (6): README.md, lobe_map.csv, requirements.txt, step47_reference.py, step49_portable.py, summary.json

Archived README synopsis: UDCT v7.14 — Step 49: OM negative-lobe map (exploratory) This is an exploratory numerical study, not preregistered and not a fit to individual Hydrus I stellar velocities . Model and methods - Plummer-generated spherical gravitational field, QUMOND-like interpolation g = 0.5 gN + sqrt(0.25 gN^2 + a0 gN) . - Tracer density rho(r) = [1+(r/a)^p]^(-5/p) . - Osipkov–Merritt augmented density rho_OM = (1+r^2/ra^2) rho . - Inversion implementation imported from the prior Step 47/48 code, included in this archive as step47_reference.py

. - Parameters: p = 1.49, 1.5, 1.501; ra/a = 0.35, 0.4, 0.45, 0.5, 0.65, 1, infinity. - Search r/a in [0.06,2], 55 log-spaced points. 96-node quadrature for brackets and 192-node quadrature for root/minimum refinement. - lobe_map.csv records bracketed roots, intervals and numerical minima. The r/a variable labels energy via turning radius; it is not a physical orbit-occupancy interval. Main observations 21 parameter pairs, 9 with detected negative intervals in the searched domain. For p=1.5: ra/a roots (r/a) min f --- --- --- 0.35 0.1665216234, 0.8545050504 -0.0020007919203 0.40 0.2237024398, 0.7669697146 -0.0008538448012 0.45 0.3381470120, 0.6206727785 -0.0001583586487 0.50 none found n/a The two roots approach each other as ra/a grows. This does not yet establish the exact double-root critical point: tangent roots can be missed by sign-change searches. A follow-up should jointly solve f=0 and df/d(log [excerpt]

Archived machine-readable result excerpt: {"p_values": [1.49, 1.5, 1.501], "ra_values": [0.35, 0.4, 0.45, 0.5, 0.65, 1.0, "inf"], "r_over_a_scan_range": [0.06, 2.0], "r_grid_points": 55, "model": "isolated spherical Plummer-generated QUMOND-like gravity, tracer [1+(r/a)^p]^(-5/p), OM", "total_parameter_pairs": 21, "pairs_with_negative_interval": 9, "quadrature_n_for_brackets": 96, "quadrature_n_for_roots_minima": 192, "limitations": ["Finite r/a domain; zero roots outside interval untested", "Shared physical inversion kernel; not independent physics check", "Near-tangent zero crossings may escape sign-change scan"]}

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

Step 50 — archived source and numerical evidence

Files (7): README.md, critical_tangencies.csv, p1p5_lobe_collapse.csv, requirements.txt, step47_reference.py, step50.py, summary.json

Archived README synopsis: UDCT v7.14 Step 50 — OM negative-lobe tangency audit Exploratory, not preregistered . Independent of observational stellar-velocity fitting. For the OM augmented density, f(Q;ra)=f_iso(Q)+B(Q)/(ra/a)^2. At a zero, (ra/a)^2=-B/f_iso. A pair of zero crossings merges where this ratio reaches a local maximum with respect to log(r/a). The radius is an energy-label turning radius, not a stellar orbit radius. The script evaluates the original Step 47 quadrature kernel, searches r/a in [0.25,0.75], and checks 96/192/384 point Gauss-Legendre quadratures. A separate scan over [0.15,0.95] tests negative-lobe widths for p=1.5. This is a local finite-domain result, not a proof of global DF nonnegativity or QUMOND failure. Run: python step50.py (requires numpy, scipy, pandas). Outputs: critical_tangencies.csv , p1p5_lobe_collapse.csv , summary.json . No network needed. The bundled step47_reference.py is the computational kernel copied from the previous step. The Step 50 numerical methods and tolerances are fully recorded in step50.py .

Archived machine-readable result excerpt: {"step": 50, "p_values": [1.49, 1.5, 1.501], "critical_tangencies": [{"p": 1.49, "critical_ra_over_a": 0.4627396446066039, "tangent_r_over_a": 0.4763697156418466, "f_at_tangent": 0.0, "df_dlogr_finite_difference": -1.0446593676910476e-09, "critical_ra_n96": 0.462739644606257, "critical_ra_at_same_r_n384": 0.4627396446052533, "delta_ra_n384_minus_n192": -1.350586309456503e-12, "domain": "r/a 0.25 to 0.75; local lobe only"}, {"p": 1.5, "critical_ra_over_a": 0.46646678739601644, "tangent_r_over_a": 0.47409755283222904, "f_at_tangent": 0.0, "df_dlogr_finite_difference": -1.0578452176723818e-09, "critical_ra_n96": 0.4664667873965174, "critical_ra_at_same_r_n384": 0.466466787395183, "delta_ra_n384_minus_n192": -8.33444424586105e-13, "domain": "r/a 0.25 to 0.75; local lobe only"}, {"p": 1.501, "critical_ra_over_a": 0.4668411650904537, "tangent_r_over_a": 0.4738661838352446, "f_at_tangent": 0.0, [excerpt]

Audit interpretation: This step is included as an original archived exploratory stage. The file inventory and reported numerical outputs do not alone establish physical validation, independent rerun, or applicability beyond the step-specific parameter window.

14. Mathematical checks needed before confirmatory interpretation

1. Re-derive the QUMOND spherical potential with consistent potential zero, boundary and deep-MOND interpolation.

2. Compute central expansion to the first nonzero term at p=1.5 and verify cancellation of all boundary contributions.

3. Implement independent right-sided Abel and conventional relative-potential inversions and compare after a sign-convention transformation.

4. Forward-integrate the recovered DF to reconstruct the input tracer density and quantify residuals over radius.

5. Test full accessible Q, including low-energy and outer-tail limits, rather than only 0.25≤r/a≤0.75.

6. Test alternative tracer laws, source masses and anisotropy families to expose model dependence.

7. Test the Hessian under mesh refinement, coordinate rotations, external-field boundaries and independently differentiated analytic benchmarks.

8. Fit individual stellar velocities and nuisance parameters using a held-out, preregistered observational protocol.

15. Discussion: what the program adds and what it does not

The research value lies in preserving a progressive computational stress test with explicit numerical failures, local parameter boundaries, code and source records. The model-specific critical-exponent argument and OM negative-lobe mapping may help refine future tests. Their novelty relative to existing phase-space consistency literature remains a bibliographic research question rather than a priority claim. This manuscript neither presents a novel complete GQUMOND action nor establishes that the MOND family fails observationally. A negative DF in a chosen model is scientifically useful because it identifies an inadmissible equilibrium construction and motivates more discriminating tests.

16. Conclusions

The two original archives document 50 exploratory stages with a scientifically meaningful separation between 3D Hessian diagnostics, spherical Jeans models and phase-space DF inversions. For the isolated Plummer source/tracer family, the central leading coefficient changes sign on either side of p=1.5 under stated assumptions; exact equality is unresolved by that term. OM scans reveal a reproducible archived local tangency around r_a/a=0.466466787396 at p=1.5 in a specified finite interval, not a global positivity boundary. Hydrus I enters as observational motivation rather than a full stellar likelihood test. A future preregistration should freeze hypotheses, numerical thresholds, datasets and independent-validation protocols before confirmatory inference.

References — external scholarly literature

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Source provenance note: prior UDCT DOI values were supplied in the author’s publication record; the present compilation has not independently verified every historical DOI landing page. Original v7.14 numerical artifacts are preserved in the companion ZIP.